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KPK BOARD TEXTBOOK NOTES - MATHEMATICS CLASS 9

Unit 12: Line Bisectors and Angle Bisectors

Complete, easy English notes based on *A Textbook of Mathematics for Grade IX*, Khyber Pakhtunkhwa Textbook Board (KPTB), Peshawar - the six theorems of the unit (12.1 to 12.6) with Given, To Prove and proof outlines, the circumcenter and the incenter, the book's worked examples, solved exercise highlights, and exam practice.

Notes by **Shah Fahad**
Primary School Teacher (PST), ESED Khyber Pakhtunkhwa - Buner, Pakistan
Contact: 03409090945 | November 2026

> **How to use these notes:** Every theorem in this unit is about one word - **equidistant** (at equal distances). Learn each pair together: a theorem and its converse. In the exam, write every proof in the order Given - To Prove - Proof, with a neat labelled figure.

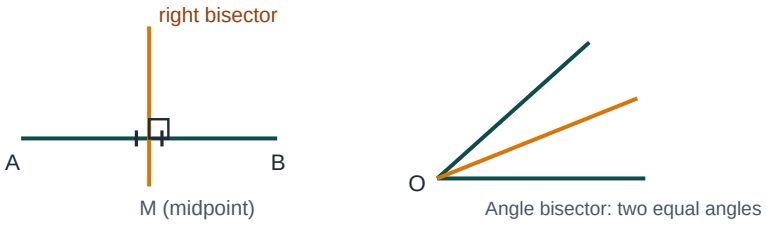
What this unit covers

1. Bisectors, midpoints and the right (perpendicular) bisector - the key definitions
2. Theorem 12.1 - a point on the right bisector is equidistant from the end points
3. Theorem 12.2 - the converse: an equidistant point lies on the right bisector
4. Theorem 12.3 - right bisectors of a triangle are concurrent: the circumcenter
5. Theorem 12.4 - a point on an angle bisector is equidistant from the arms
6. Theorem 12.5 - the converse for angle bisectors
7. Theorem 12.6 - angle bisectors of a triangle are concurrent: the incenter
8. Solved exercise highlights, summary, MCQs and practice questions with answers

Source book: KPTB Mathematics for Grade IX, Unit 12 (book pages 256-266). The notes follow the book's theorems, examples and exercises in order. The scan used for verification carries no printed copyright page, so no printing year is claimed here - the 17-unit structure is the current circulating KPTB edition.

1. Bisectors - The Key Definitions

Term	Definition (as given in the book)
Segment bisector	A line that divides a segment into two congruent parts.
Midpoint	The point where a bisector intersects a segment is the midpoint of the segment.
Perpendicular bisector (right bisector)	A bisector that is perpendicular to a segment. The book also calls it the **right bisector** of the segment.
Angle bisector	A line or ray that divides an angle into two congruent angles.
Equidistant	At equal distances. A point is equidistant from two points when its distances to them are equal, and equidistant from the arms of an angle when its *perpendicular* distances to the two arms are equal.
Concurrent	Lines are concurrent when they all pass through one single point. That point is called the point of concurrency.



Left: the right bisector crosses segment AB at its midpoint M at 90 degrees. Right: a ray from O splitting an angle into two congruent angles.

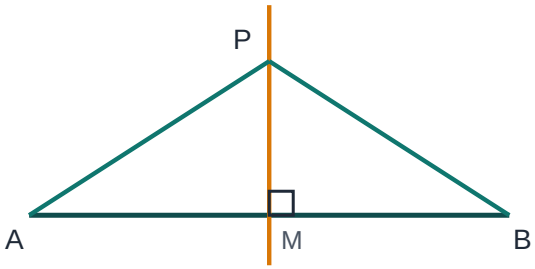
> **The one idea of the unit:** a right bisector is the set of *all* points equidistant from the end points of a segment, and an angle bisector is the set of *all* points equidistant from the arms of the angle. Theorems 12.1 and 12.4 prove one direction; Theorems 12.2 and 12.5 prove the way back.

Notation used in this unit

- A bar over two letters, such as AB with a bar, means the line segment from A to B; the same letters without a bar mean its length.
- The symbol for congruent is used for segments and angles of equal measure.
- "Perpendicular distance" from a point to a line is always measured along the perpendicular from the point to that line - never along a slanted line.

2. Theorem 12.1 - Points on the Right Bisector

Theorem 12.1. Any point on the right bisector of a line segment is equidistant from its end points.



PM is the right bisector of AB, so M is the midpoint and the angles at M are right angles. The theorem says PA = PB.

Given: P is a point on the right bisector PM of segment AB; M is the midpoint of AB. **To Prove:** PA is congruent to PB.

Proof outline (the book's proof)

Statements	Reasons
In triangles PMA and PMB: AM is congruent to BM.	M is the midpoint of AB
Angle PMA = angle PMB = 90 degrees.	PM is the right bisector of AB
PM is common to both triangles, so triangle PMA is congruent to triangle PMB.	SAS congruence
Therefore PA is congruent to PB.	Corresponding parts of congruent triangles

How it is used: whenever a point sits on a perpendicular bisector, immediately write the two distances equal. This single step solves most numerical questions in the unit - see the solved Exercise 12.1 on page 10, where AB is set equal to BC again and again.

3. Theorem 12.2 - The Converse of 12.1

Theorem 12.2. Any point equidistant from the end points of a line segment is on the right bisector of it.

Given: P is a point with PA congruent to PB. M is the midpoint of AB, and P is joined to M. **To Prove:** PM is the right bisector of AB - that is, PM is perpendicular to AB at M.

Proof outline (the book's proof)

Statements	Reasons
In triangles PMA and PMB: PA is congruent to PB, and AM is congruent to BM.	Given; M is the midpoint
PM is common, so triangle PMA is congruent to triangle PMB.	SSS congruence
Therefore angle PMA = angle PMB.	Corresponding parts of congruent triangles
But angles PMA and PMB form a linear pair, so their sum is 180 degrees; equal angles summing to 180 degrees are each 90 degrees.	Supplementary angles
So PM is perpendicular to AB at its midpoint - PM is the right bisector of AB, and P lies on it.	Definition of a right bisector

Example 1 (the book's example). For what value of x does the point P lie on the right bisector of segment AB, when the two distances from P to the end points are given in terms of x?

Solution: By Theorem 12.2, P lies on the bisector exactly when the distances are equal. Setting the two segment lengths equal and solving gives **x = 5**. The method is always the same: *set the distances equal, then solve*.

Do not swap 12.1 and 12.2. Theorem 12.1 starts *on* the bisector and concludes the distances are equal. Theorem 12.2 starts *from* equal distances and concludes the point is on the bisector. Quote the right one in a proof.

4. Theorem 12.3 - The Circumcenter of a Triangle

Theorem 12.3. The right bisectors of the sides of a triangle are concurrent.

Given: In triangle ABC, the right bisectors of sides BC and CA meet at point O. O is joined to A, B and C. **To Prove:** the right bisector of the third side AB also passes through O.

Proof outline (the book's proof)

Statements	Reasons
O lies on the right bisector of BC, so OB is congruent to OC.	Theorem 12.1
O lies on the right bisector of CA, so OC is congruent to OA.	Theorem 12.1
Therefore OA is congruent to OB.	From the two steps above
So O, being equidistant from A and B, lies on the right bisector of AB as well.	Theorem 12.2

The circumcenter and the circumcircle

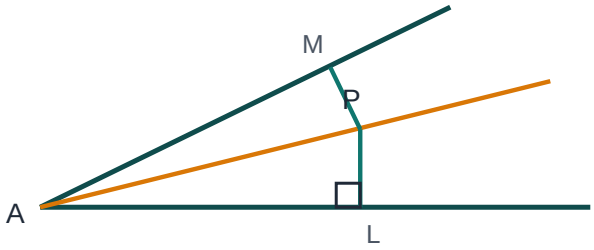
The point of concurrency of the three right bisectors is called the **circumcenter** of the triangle. The proof shows it is **equidistant from the three vertices**: OA = OB = OC.

Because of this, a circle drawn with the circumcenter as centre and the distance to any one vertex as radius passes through all three vertices. This circle is said to be **circumscribed** about the triangle, and it is called the **circumcircle**.

> **Position:** unlike the incenter (page 8), the circumcenter is not always inside the triangle. In an acute-angled triangle it lies inside; in a right-angled triangle it is the midpoint of the hypotenuse; in an obtuse-angled triangle it falls outside.

5. Theorem 12.4 - Points on an Angle Bisector

Theorem 12.4. Any point on the bisector of an angle is equidistant from its arms.



P is on the bisector of angle A. PL and PM are the perpendiculars from P to the two arms. The theorem says PL = PM.

Given: AP bisects angle A, P is a point on AP, and PL and PM are perpendicular to the two arms of the angle. **To Prove:** PL is congruent to PM.

Proof outline (the book's proof)

Statements	Reasons
In triangles PLA and PMA: angle PLA = angle PMA = 90 degrees.	PL and PM are perpendicular to the arms
Angle LAP = angle MAP, and AP is common to both triangles.	AP bisects angle A; common side
So triangle PLA is congruent to triangle PMA.	SAA congruence
Therefore PL is congruent to PM.	Corresponding parts of congruent triangles

> **Remember:** "equidistant from the arms" always means the **perpendicular** distances PL and PM. A slanted distance from P to an arm does not count, however equal it may look.

6. Theorem 12.5 - The Converse of 12.4

Theorem 12.5. If a point is equidistant from the arms of an angle, then it lies on the bisector of the angle.

Given: Point P inside angle A has PL congruent to PM, where PL and PM are perpendicular to the arms of the angle. **To Prove:** AP is the bisector of angle A, that is, angle LAP = angle MAP.

Proof outline (the book's proof)

Statements	Reasons
In right triangles PLA and PMA: PL is congruent to PM.	Given - P is equidistant from the arms
AP is the common hypotenuse of both right triangles, so the triangles are congruent.	HL (right angle - hypotenuse - side) congruence
Therefore angle LAP = angle MAP.	Corresponding parts of congruent triangles
So AP bisects angle A, and P lies on its bisector.	Definition of an angle bisector

Example 2 (the book's example). For what value of x does P lie on the bisector of angle A, when the distances from P to the two sides of the angle are labelled x + 3 and 2x - 1?

Solution: By the converse of the angle bisector theorem (12.5), P lies on the bisector when P is equidistant from the sides, so set the lengths equal: x + 3 = 2x - 1, which gives **x = 4**.

Pair them up: 12.1 and 12.2 are a theorem-and-converse pair about *end points of a segment*; 12.4 and 12.5 are the matching pair about *arms of an angle*. The two concurrency theorems, 12.3 and 12.6, are built by using each pair twice.

7. Theorem 12.6 - The Incenter of a Triangle

Theorem 12.6. The bisectors of the angles of a triangle are concurrent.

Given: BD and CD are the bisectors of angles B and C of triangle ABC, meeting at D; D is joined to A. From D, perpendiculars are drawn to the three sides. **To Prove:** DA is the bisector of angle A.

Proof outline (the book's proof)

Statements	Reasons
D lies on the bisector of angle B, so D is equidistant from the arms of angle B: its perpendicular distances to BC and to BA are equal.	Theorem 12.4
D lies on the bisector of angle C, so its perpendicular distances to BC and to CA are also equal.	Theorem 12.4
Therefore the distances from D to BA and to CA are equal - D is equidistant from the arms of angle A.	From the two steps above
So D lies on the bisector of angle A: all three bisectors pass through D.	Theorem 12.5

The incenter and the incircle

The point of concurrency of the three angle bisectors is called the **incenter** of the triangle. The incenter **always lies inside** the triangle, and it is equidistant from the three sides.

A circle drawn with the incenter as centre and the distance to any one side as radius touches all three sides from inside. It is said to be **inscribed** within the triangle, and is called the **incircle**.

> **Book definition:** an angle bisector of a triangle is the bisector of the interior angle of the triangle - so each bisector starts at a vertex and ends on the opposite side, inside the triangle.

8. Circumcenter and Incenter - Side by Side

Feature	Circumcenter	Incenter
Formed by	Right bisectors of the three sides (Theorem 12.3)	Bisectors of the three angles (Theorem 12.6)
Equidistant from	The three **vertices**	The three **sides**
Circle	Circumcircle - passes through all three vertices; triangle is inside it	Incircle - touches all three sides; circle is inside the triangle
Radius	Distance from the centre to any vertex	Perpendicular distance from the centre to any side
Position	Inside, on the hypotenuse, or outside, according to the angles	Always inside the triangle

Special triangles

- **Isosceles triangle:** the right bisector of the base passes through the vertex angle opposite the base (Review Exercise 12, MCQ iv). The same line is also the bisector of the vertex angle, the median to the base and the altitude to the base - four lines in one.
- **Equilateral triangle:** the perpendicular bisectors are also the angle bisectors (Review Exercise 12, MCQ vii) - and also the medians and altitudes. The circumcenter and incenter are the *same point*.
- **Base angles of an isosceles triangle:** the bisectors of the two base angles meet each other on the right bisector of the base (Review Exercise 12, Question 4) - because their meeting point is equidistant from the base's end points.

Example. The circumcenter O of triangle ABC has OA = 5 cm. What are OB and OC, and what circle can be drawn?

Solution: The circumcenter is equidistant from the vertices, so OB = OC = **5 cm**. A circle centred at O with radius 5 cm passes through A, B and C - the circumcircle of the triangle.

Example. The incenter I of a triangle is 3 cm (perpendicular distance) from side BC. How far is it from the other two sides?

Solution: The incenter is equidistant from all three sides, so it is also **3 cm** from each of the other sides, and the incircle has radius 3 cm.

9. Solved Exercise Highlights (Book Exercise 12.1)

Question	Answer
BD is the right bisector of AC in the figure; AB is labelled 5x and BC is labelled 4x + 3. Find the length of AB. (Q4)	B is on the right bisector, so AB = BC (Thm 12.1): 5x = 4x + 3, x = 3, AB = 15
BD is the perpendicular bisector of AC; AD = 20 and DC = 7x - 15, BC = 6x - 5. (i) Which segment lengths are equal? (ii) Find x. (iii) Find AB. (Q5)	(i) AD = DC and AB = BC (ii) 20 = 7x - 15, so x = 5 (iii) AB = BC = 6(5) - 5 = 25
Find x: (i) a point on an angle bisector with distances x + 11 and 3x + 1 to the arms. (Q7i)	Distances equal (Thm 12.4): x + 11 = 3x + 1, so x = 5
Find x: (ii) a bisector splitting an angle into 7x degrees and (3x + 16) degrees. (Q7ii)	Parts equal: 7x = 3x + 16, so x = 4
Find x: (iii) a bisector splitting an angle into (5x - 2) degrees and (3x + 14) degrees. (Q7iii)	Parts equal: 5x - 2 = 3x + 14, so x = 8

Proof questions set in the book - know the plan for each

- **Q1:** If the diagonals of a quadrilateral are the right bisectors of each other, all four sides are congruent. Plan: each vertex lies on the right bisector of a diagonal, so apply Theorem 12.1 at each corner - the four sides come out equal in turn (the quadrilateral is a rhombus).
- **Q2 and Q3:** congruence practice built on this unit's figures - prove the two named triangles congruent (watch for vertical angles, common sides and the equal parts a bisector gives), then read off the required result.
- **Q6:** Can we conclude that EH bisects angle FEG? Use Theorem 12.5: the conclusion follows exactly when H is equidistant from the arms EF and EG, measured by perpendiculars.
- **Q8:** The diagonals of a square are the right bisectors of each other. Plan: a square is a parallelogram (diagonals bisect each other, Unit 11) and its diagonals are perpendicular - both halves of the definition are met.
- **Review Q2:** AB = AD and CB = CD implies AC is perpendicular to BD and BE = DE - because both A and C lie on the right bisector of BD (Theorem 12.2).
- **Review Q3:** The diagonals of a rhombus are the right bisectors of each other - same plan as Q8.

10. Unit Summary and Common Mistakes

The whole unit on one page

- **Definitions:** a segment bisector divides a segment into two congruent parts at its midpoint; a perpendicular (right) bisector does so at 90 degrees; an angle bisector divides an angle into two congruent angles.
- **12.1:** Any point on the right bisector of a segment is equidistant from its end points.
- **12.2:** Any point equidistant from the end points of a segment is on the right bisector of it.
- **12.3:** The right bisectors of the sides of a triangle are concurrent - at the circumcenter, equidistant from the vertices (circumcircle).
- **12.4:** Any point on the bisector of an angle is equidistant from its arms.
- **12.5:** A point equidistant from the arms of an angle lies on the bisector of the angle.
- **12.6:** The bisectors of the angles of a triangle are concurrent - at the incenter, equidistant from the sides (incircle), always inside the triangle.
- **Numerical rule:** in every find-x question, set the two equal things equal - distances for a bisector, angle parts for an angle bisector - then solve.

Common mistakes

- **End points vs arms:** right bisector goes with *end points* of a segment; angle bisector goes with *arms* of an angle. Swapping them is the most common lost mark.
- **Slanted distances:** distance to an arm or a side means the perpendicular distance only.
- **Bisector vs right bisector:** a plain bisector need not be perpendicular. The equidistance theorems need the *right* bisector.
- **Centres mixed up:** circumcenter - vertices and circumcircle; incenter - sides and incircle. Also remember only the incenter is guaranteed to be inside.
- **Quoting the converse by mistake:** check which direction your Given and To Prove need before writing the theorem number.

11. Practice - MCQs and Questions (Book Review Exercise 12)

1. Which of the following are concurrent? (a) Angle bisectors of a triangle (b) Perpendicular bisectors of the sides of a triangle (c) Medians of a triangle (d) All of these
2. Which of the following is sometimes inside a triangle, sometimes coincides with a side, and sometimes falls outside? (a) The base (b) The altitude (c) The median (d) The angle bisector
3. Perpendicular bisectors of a triangle are: (a) Congruent (b) Concurrent (c) Parallel to each other (d) Perpendicular to each other
4. In which triangle does the perpendicular bisector of the base pass through its vertex angle? (a) Right angled (b) Scalene (c) Isosceles (d) Acute-angled
5. In triangle ABC, medians AD, BE and CF intersect at G. If CF = 24, what is the length of FG? (a) 8 (b) 12 (c) 10 (d) 16
6. The angle bisectors of a triangle meet at a point which is equidistant from ..... of the triangle. (a) The vertices (b) The sides (c) Midpoints of the sides (d) All of these
7. In an equilateral triangle, all the perpendicular bisectors are: (a) Congruent (b) Concurrent (c) The angle bisectors as well (d) Parallel
8. The point of intersection of the angle bisectors of a triangle is equidistant from ..... of the triangle. (a) The vertices (b) The sides (c) Midpoints of the sides (d) All of these

Short questions

1. A point Q on the right bisector of MN has QM = 9 cm. Find QN, and name the theorem you used.
2. A point on the bisector of a 64-degree angle: what is each part of the angle, and what can you say about the point's distances to the arms?
3. Prove that the diagonals of a rhombus are the right bisectors of each other. (Review Q3)
4. State Theorem 12.3 and Theorem 12.6, naming the two centres they define.

Answer key - MCQs: 1 (d) 2 (b) 3 (b) 4 (c) 5 (a) - the centroid divides the median 2 : 1, so FG = 24 / 3 = 8 6 (b) 7 (c) - in an equilateral triangle each perpendicular bisector is also an angle bisector (they are concurrent too, as in every triangle) 8 (b).
Short questions: 1. QN = 9 cm, by Theorem 12.1. 2. Each part is 32 degrees; the perpendicular distances to the two arms are equal (Theorem 12.4). 3. Rhombus diagonals bisect each other (Unit 11) and are perpendicular - the two halves of the right-bisector definition. 4. 12.3 defines the circumcenter; 12.6 defines the incenter.